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Architecture Design Document

3-Tier DevOps Quiz Application on AWS EKS

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1. Executive Summary

This document describes the architecture of a production-ready 3-tier web application deployed on AWS EKS. The system consists of a React frontend, Flask REST API backend, and PostgreSQL database, designed with AWS Well-Architected Framework principles and regulatory compliance (GDPR, SOC2, EU/USA regulations).

2. Architecture Diagram

```
Parse error on line 2:
...ph TB      subgraph "Internet"          US
-----^
Expecting 'SEMI', 'NEWLINE', 'SPACE', 'EOF', 'GRAPH',
'DIR', 'TAGEND', 'TAGSTART', 'UP', 'DOWN', 'subgraph',
'end', 'SQE', 'PE', '(-)', 'DIAMOND_STOP', 'MINUS', '--',
'ARROW_POINT', 'ARROW_CIRCLE', 'ARROW_CROSS', 'ARROW_OPEN',
'DOTTED_ARROW_POINT', 'DOTTED_ARROW_CIRCLE',
'DOTTED_ARROW_CROSS', 'DOTTED_ARROW_OPEN', '==',
'THICK_ARROW_POINT', 'THICK_ARROW_CIRCLE',
'THICK_ARROW_CROSS', 'THICK_ARROW_OPEN', 'PIPE', 'STYLE',
'LINKSTYLE', 'CLASSDEF', 'CLASS', 'CLICK', 'DEFAULT',
'NUM', 'PCT', 'COMMA', 'ALPHA', 'COLON', 'BRKT', 'DOT',
'PUNCTUATION', 'UNICODE_TEXT', 'PLUS', 'EQUALS', 'MULT',
got 'STR'
```

3. Well-Architected Framework Alignment

3.1 Security Pillar

Control	Implementation
Encryption at Rest	KMS for RDS, EKS secrets, EBS volumes, S3 buckets
Encryption in Transit	TLS via ACM + ALB, inter-service HTTPS
Identity & Access	IRSA for service accounts, OIDC for CI/CD
Network Security	Private subnets, Security Groups, Network Policies
Secrets Management	AWS Secrets Manager with KMS encryption
Audit Logging	CloudTrail (multi-region), VPC Flow Logs
Threat Detection	GuardDuty with K8s audit logs
Key Rotation	Automatic KMS key rotation enabled

3.2 Reliability Pillar

Control	Implementation
Multi-AZ	RDS Multi-AZ (prod), EKS across 2 AZs
Auto-scaling	HPA (CPU 50%, Memory 70%), 2-5 replicas
Pod Disruption	PDB with minAvailable=1
Health Checks	Liveness + Readiness probes on all pods
Backup & Recovery	7-day RDS retention, cross-region snapshots
DR Strategy	RPO < 1hr, RTO < 4hr (see Section 6)

3.3 Operational Excellence

Control	Implementation
IaC	100% Terraform with modules
Monitoring	Prometheus + Grafana stack
Alerting	CloudWatch alarms (RDS CPU, storage)
CI/CD	GitHub Actions with OIDC (no static keys)
Logging	EKS control plane logs (90-day retention)

3.4 Cost Optimization

Control	Implementation
Right-sizing	t3.medium nodes, db.t3.micro RDS
Auto-scaling	HPA prevents over-provisioning
Storage	gp3 volumes (20% cheaper than gp2)
Environment-aware	Conditional Multi-AZ, deletion protection

3.5 Performance Efficiency

Control	Implementation
Caching	Nginx static file serving with try_files

Control	Implementation
Load Balancing	ALB with path-based routing
Resource Limits	CPU/memory requests and limits set
Performance Insights	RDS Performance Insights enabled

4. Compliance Matrix

Regulation	Requirement	Implementation
GDPR	Data encryption	KMS encryption at rest, TLS in transit
GDPR	Audit trail	CloudTrail with 365-day retention
GDPR	Data residency	Configurable region deployment
GDPR	Right to erasure	Database delete APIs available
SOC2	Access control	RBAC, least-privilege IAM policies
SOC2	Monitoring	GuardDuty, CloudWatch, Prometheus
SOC2	Change management	Terraform IaC, Git version control
SOC2	Incident response	SNS alerting, CloudWatch alarms
EU/USA	Network security	VPC isolation, Security Groups, Network Policies
EU/USA	Configuration compliance	AWS Config recorder

5. Network Architecture

Subnet	CIDR	Purpose	Access
Public-1	10.0.1.0/24	ALB, NAT Gateway	Internet-facing
Public-2	10.0.2.0/24	ALB (Multi-AZ)	Internet-facing
Private-1	10.0.3.0/24	EKS Worker Nodes	Via NAT Gateway
Private-2	10.0.4.0/24	EKS Worker Nodes (Multi-AZ)	Via NAT Gateway

Subnet	CIDR	Purpose	Access
RDS-1	10.0.5.0/24	PostgreSQL RDS	EKS subnets only
RDS-2	10.0.6.0/24	PostgreSQL RDS (Multi-AZ)	EKS subnets only

6. Disaster Recovery Strategy

Parameter	Value
DR Tier	Pilot Light
RPO	< 1 hour (automated RDS backups)
RTO	< 4 hours (Terraform re-deploy)
Primary Region	ap-south-1 (Mumbai)
DR Region	us-east-1 (N. Virginia)

Recovery Procedures

- Database:** Restore from latest automated snapshot or cross-region copy
- EKS:** Re-deploy via Terraform in DR region
- Application:** Re-deploy K8s manifests, update DNS in Route53
- Failback:** Reverse replication after primary recovery

7. Data Flow

